

Bio-Inspired Routing Optimization in Social Wasps

Project Overview

Bio-Inspired Routing Optimization in Social Wasps is an agent-based simulation project that investigates how different movement and routing strategies affect the efficiency of larval feeding in social wasp colonies.

The project transforms observed social-wasp nest and feeding-bout data into a controlled computational benchmark. In the simulated environment, larvae remain fixed within a nest while adult wasps move through a two-dimensional grid to locate and feed hungry larvae.

The central research question is:

How do different wasp movement strategies perform when feeding the same synthetic larval colony under identical environmental and simulation conditions?

Four movement strategies are evaluated:

- Random: A purely random movement strategy used as a baseline.
- Biased: A directionally persistent movement strategy that reduces completely random wandering.
- Greedy: A local optimization strategy that prioritizes nearby, hungry, higher-stage larvae while controlling excessive target crowding.
- TSP: A structured route-planning strategy based on nearest-neighbor routing across remaining unfed larvae.

The objective is not to claim that real wasps solve the Traveling Salesperson Problem. Instead, the TSP strategy provides a structured routing benchmark against which simpler movement strategies can be compared.

Motivation

Larval feeding in social wasps can be viewed as both a biological behavior and a spatial optimization problem.

Larvae are distributed across different locations within the nest and may differ in developmental stage and hunger level. Adult wasps must repeatedly move through the nest, locate appropriate larvae, and deliver food. Inefficient movement can result in unnecessary travel and slower colony-wide feeding.

This creates an interesting computational problem:

Can structured movement strategies reduce the time and distance required to feed an entire colony?

The project uses real nest geometry and behavioral observations as the foundation for the simulation while keeping the actual decision-making strategies under controlled experimental conditions.

This separation allows the project to investigate routing performance without simply reproducing the observed feeding behavior.

Research Question

The primary research question is:

How do different wasp movement strategies compare in feeding the same synthetic larval colony in terms of completion time, total movement, movement efficiency, and final larval hunger?

The project specifically investigates whether structured route planning provides measurable advantages over local, directional, and random movement.

Project Objectives

The main objectives of the project are:

- Convert observed wasp nest data into computational simulation environments.
- Reconstruct nest geometry using a two-dimensional grid.
- Generate reproducible synthetic larval colonies from observed nest structures.
- Assign biologically interpretable initial hunger values based on larval developmental stage.
- Implement multiple wasp movement strategies.
- Run every strategy under equivalent benchmark conditions.
- Measure routing performance using quantitative metrics.
- Compare strategy performance across all available nest-bout scenarios.
- Produce visual simulations that make the routing behavior interpretable.
- Provide a reproducible computational pipeline through Python and Jupyter Notebook.

Dataset

The project uses two locally stored CSV datasets.

Nest-Cell Dataset

File: ED_FL_3nests1noC2.csv

This dataset provides the spatial and biological structure of the nests, including:

- Nest identification
- Cell coordinates
- Cell contents
- Larval developmental stage
- Distance from nest center

Feeding-Bout Dataset

File: ALL_FL_minmaj_final3noC2.csv

This dataset contains behavioral observations associated with feeding bouts, including:

- Nest ID
- Bout ID
- Behavior code
- Cell reference
- Wasp ID
- Feeding activity information

The original datasets are kept locally and are not included in the repository.

Simulation Model

The simulation is implemented using the Mesa agent-based modeling framework in Python.

The nest is represented as a two-dimensional grid. Each occupied larval cell contains a stationary larval agent, while adult wasps are represented as mobile agents.

At every simulation step, wasps:

1. Determine their next movement according to the selected strategy.
2. Move through the nest grid.
3. Detect larvae at their current location.
4. Attempt to feed eligible larvae.
5. Continue moving until the colony reaches the required feeding condition.

This agent-based structure allows different movement rules to operate within exactly the same simulated environment.

Synthetic Colony Construction

To create a sufficiently challenging routing benchmark, the observed larval populations are scaled by a factor of two.

The benchmark uses:

Scale factor = 2

The original larval layouts are duplicated using deterministic coordinate jitter. This preserves the general spatial organization of the observed nest while producing a larger synthetic colony for stress testing.

The resulting benchmark contains:

Nest	Original Larvae	Scaled Larvae	Bouts
v14	34	68	10
v72	53	106	14
v87	67	134	12

This produces:

3 nests → 36 nest-bout scenarios → 144 strategy simulations

Each of the 36 scenarios is evaluated using all four movement strategies.

Hunger Initialization

An important design decision is that simulated hunger is not directly copied from observed feeding frequency.

Instead, hunger is randomly initialized according to larval developmental stage.

Larval Stage	Initial Hunger Range
L1	0.20 – 0.50
L2	0.45 – 0.75
L3	0.65 – 1.00

This approach avoids circular reasoning.

Observed feeding frequency is retained as behavioral metadata, while simulated hunger is independently generated using biologically interpretable developmental-stage ranges.

The benchmark therefore tests the movement strategies rather than simply reproducing the observed feeding-frequency distribution.

Experimental Conditions

To ensure a fair comparison, the simulations use fixed experimental conditions.

Parameter	Value
Scale factor	2
Random seed	42
Benchmark horizon	3000 steps
Number of strategies	4
Number of scenarios	36

The number of wasps is determined using the scenario's scaled larval population and observed behavioral activity:

$$\text{Wasps} = \max(\text{ceil}(\text{scaled larvae} / 5), \text{observed unique wasps}) + \text{activity bonus}$$

where:

$$\text{Activity bonus} = \text{ceil}(\text{observed feeding events} / 100)$$

Each strategy receives the same scenario configuration so that differences in performance are attributable to the movement rule rather than differences in colony size, grid structure, or available resources.

Movement Strategies

1. Random Strategy

The random strategy provides the baseline control.

Wasps move without considering larval hunger, developmental stage, or the global distribution of targets.

This strategy represents inefficient exploratory movement and provides a useful lower-bound comparison for the more informed strategies.

2. Biased Strategy

The biased strategy introduces directional persistence into movement.

Instead of choosing every movement completely independently, the wasp tends to continue moving in a similar direction.

This reduces some of the chaotic behavior of the random strategy but does not explicitly optimize the location of hungry larvae.

3. Greedy Strategy

The greedy strategy uses local information to select promising feeding targets.

Target selection considers factors such as:

- Larval hunger
- Distance from the wasp
- Larval developmental stage
- Existing target crowding

The goal is to obtain a useful local optimization strategy without requiring a complete global route.

This makes greedy routing a meaningful comparison against the globally organized TSP-style strategy.

4. TSP Strategy

The TSP strategy is implemented as a nearest-neighbor route-style benchmark.

The strategy considers remaining unfed larvae and repeatedly constructs a route toward appropriate targets.

It is designed to test the value of route-level organization compared with local decision-making.

Importantly, this strategy should not be interpreted as a biological claim that social wasps explicitly solve the Traveling Salesperson Problem.

Instead:

TSP is used as an optimization benchmark for measuring the potential benefit of organized spatial routing.

Performance Metrics

Each simulation is evaluated using multiple performance measures.

Completion Time

The number of simulation steps required to complete feeding across the colony.

Lower values indicate faster colony-level feeding.

Total Movement Distance

The total distance traveled by the wasps during the simulation.

Lower values indicate less wasted movement.

Final Average Hunger

The average hunger remaining across larvae at the end of the simulation.

Lower values indicate more effective feeding.

Movement Efficiency

A combined measure of feeding progress relative to movement expenditure.

Higher values indicate that the strategy obtains greater feeding progress for the amount of movement performed.

Using multiple metrics prevents the analysis from defining "best" purely as the fastest strategy.

Benchmark Results

The final benchmark produces the following results:

Strategy	Runs	Finish Rate	Median Completion Step	Median Distance	Median Final Hunger	Median Efficiency
TSP	36	100%	67.0	2,345.0	0.017	45.1
Greedy	36	100%	142.5	6,137.0	0.118	17.3
Biased	36	100%	463.5	8,353.5	0.286	12.5
Random	36	100%	529.0	11,642.5	0.209	8.8

All four strategies eventually completed the benchmark scenarios, but their efficiency differed substantially.

The TSP strategy achieved the strongest overall performance across the measured metrics.

Results Interpretation

The TSP strategy completed the scenarios considerably faster than the other movement strategies.

Its median completion time was:

67 steps

compared with:

- 142.5 steps for greedy
- 463.5 steps for biased
- 529 steps for random

TSP also required substantially less movement.

Its median travel distance was approximately:

2,345 grid units

while the random strategy required approximately:

11,642.5 grid units

This indicates that the advantage of structured routing is not simply faster movement. The TSP strategy also avoids a large amount of unnecessary travel.

The final hunger results show a similar pattern. TSP produced a median final hunger of only 0.017, compared with 0.118 for greedy, 0.286 for biased, and 0.209 for random.

Overall, the benchmark indicates that global route organization can substantially improve feeding efficiency compared with local or uninformed movement rules.

Strategy Ranking

Based on the benchmark results, the overall strategy ranking is:

1. TSP
2. Greedy
3. Biased
4. Random

TSP

TSP provides the best combination of speed, low movement, and low remaining hunger.

Greedy

Greedy performs substantially better than random or biased movement. Its ability to prioritize useful local targets makes it a strong second-place strategy.

Biased

Directional persistence improves over completely random exploration, but without explicit target selection it still produces substantial unnecessary movement.

Random

Random movement performs worst in terms of routing efficiency because it spends a large amount of movement exploring areas that do not directly contribute to feeding.

Experimental Coverage

A major strength of the project is that the analysis is not based on a single hand-selected simulation.

The complete benchmark evaluates:

- 3 observed nests
- 36 nest-bout scenarios
- 4 movement strategies
- 144 total simulations

The final validation confirms:

- Total nests analyzed: 3
- Nests: v14, v72, v87
- Total scenarios: 36
- Total fair simulations: 144
- Best overall strategy: TSP
- Incomplete simulations after extension: 0

This broad scenario coverage makes the result more meaningful than a demonstration based on a single example.

Visualization and Simulation Output

The project also provides visual evidence of the simulation behavior.

Four playable strategy simulations are generated for the large v87 scenario:

- TSP
- Greedy
- Biased
- Random

The simulations visualize:

- Larval locations
- Wasp positions
- Nest structure
- Blocked cells
- Feeding activity
- Final larval states

The project exports both GIF animations and standalone HTML animations.

The GIFs provide easy playback in GitHub, while the HTML versions allow interactive exploration in a browser.

These visualizations complement the numerical benchmark by demonstrating that the model is actually performing spatial movement and feeding rather than only producing summary statistics.

Analysis and Visualization

The final analysis notebook focuses on visualizations that directly support the research question.

The main analysis figures include:

Strategy Scorecard

Compares speed, movement, efficiency, and final hunger across all strategies.

Scenario Heatmap

Shows completion time for every nest-bout scenario and movement strategy.

TSP Delay Analysis

Measures how much slower each alternative strategy is compared with TSP on the same scenario.

Nest-Level Comparison

Tests whether the overall strategy ranking remains consistent across nests v14, v72, and v87.

Efficiency Frontier

Visualizes the relationship between completion speed and movement efficiency.

Bout Difficulty Analysis

Examines how observed activity, spatial breadth, and colony size relate to simulated completion time.

Only plots that provide evidence for a specific research claim are retained in the final analysis.

Software Architecture

The project consists of several main components.

Data Processing Layer

Reads and processes the observed nest and behavioral CSV datasets.

Scenario Generation Layer

Combines nest geometry and bout-level observations to construct reproducible simulation scenarios.

Agent-Based Simulation Layer

Uses Mesa to represent larvae and wasps as agents operating within the nest grid.

Strategy Layer

Implements the four movement strategies:

random, biased, greedy, and tsp.

Benchmark Layer

Runs every strategy against every scenario and calculates performance metrics.

Analysis Layer

Aggregates simulation results and generates statistical summaries and visualizations.

Visualization Layer

Produces figures, GIF animations, and interactive HTML simulation outputs.

This separation makes it possible to modify a routing strategy without redesigning the entire simulation pipeline.

Technology Stack

The project is primarily implemented using:

- Python — Core programming language
- Mesa — Agent-based modeling framework
- Jupyter Notebook — Interactive analysis and reporting
- Matplotlib — Data visualization and animation
- Pandas — Data processing and analysis
- NumPy — Numerical computation
- HTML/GIF — Simulation visualization and presentation

Reproducibility

Reproducibility is a central design principle of the project.

The benchmark uses a fixed random seed:

```
RANDOM_SEED = 42
```

The same scenario configuration is provided to each strategy, allowing the strategies to be compared under controlled conditions.

The project provides both:

- A standalone Python benchmark script
- An executed Jupyter analysis notebook

The benchmark can therefore be reproduced without manually recreating the analysis.

Repository Structure

The main project files are organized as follows:

File / Directory	Purpose
final_analysis.ipynb	Main research analysis and results
wasp_routing_analysis.py	Reproducible benchmark script
generate_strategy_animations.py	Animation-generation script
figures/	Exported analysis and simulation figures
animations/	GIF and HTML simulation animations
requirements.txt	Python dependencies

`.gitignore`

Prevents local datasets and generated files from being committed

The final notebook is the primary deliverable for analysis, presentation, and project evaluation.

Running the Project

The project can be executed locally using Python.

First, clone the repository and enter the project directory.

```
git clone
https://github.com/hemu77/Bio-Inspired-Routing-Optimization-in-Social-Wasps.git
cd Bio-Inspired-Routing-Optimization-in-Social-Wasps
```

Place the two required datasets in the project root:

```
ED_FL_3nests1noC2.csv
ALL_FL_minmaj_final3noC2.csv
```

Install the required dependencies:

```
python -m pip install -r requirements.txt
```

Run the benchmark:

```
python wasp_routing_analysis.py --output-dir outputs
```

Generate the strategy animations:

```
python generate_strategy_animations.py --output-dir animations
```

The final notebook can be opened using:

```
jupyter lab final_analysis.ipynb
```

Validation

The project includes automated validation checks to ensure that the benchmark structure is correct.

The final analysis verifies:

- Three nests are included.
- All expected nest IDs are present.
- Thirty-six nest-bout scenarios are generated.
- Four strategies are executed for every scenario.
- A total of 144 simulations are completed.
- No simulations remain incomplete after the benchmark extension.
- The same overall strategy ranking is reproduced by the notebook and standalone script.

These checks provide a basic regression layer against accidental changes to the experimental pipeline.

Limitations

Although the benchmark produces clear differences between routing strategies, the simulation should not be interpreted as a complete biological reconstruction of wasp behavior.

Simplified Hunger Dynamics

Hunger growth and feeding effects are represented using stylized computational rules rather than continuous physiological measurements.

Grid-Based Movement

The nest is represented using a discrete grid. Real wasps move through continuous physical space.

Simplified Wasp Roles

Wasp behavior is simplified into computational roles such as forager, unloader, and feeder.

TSP as an Optimization Benchmark

The TSP strategy is intentionally a computational benchmark and should not be interpreted as evidence that real wasps use explicit TSP algorithms.

Single Random Seed

The final benchmark currently uses one primary random seed.

Consequently, the reported results demonstrate the behavior of the benchmark under the selected conditions but do not yet provide confidence intervals across multiple stochastic realizations.

Future Improvements

Several extensions could make the model more biologically realistic and statistically robust.

Multi-Seed Evaluation

Run each scenario across multiple random seeds and report confidence intervals or distributions rather than relying on a single seed.

Biological Calibration

Use observed feeding-bout timing to calibrate hunger growth, feeding effects, and behavioral transitions.

Additional Routing Strategies

Introduce more biologically plausible heuristics, including pheromone-inspired routing, swarm-based search, reinforcement learning, and decentralized cooperative strategies.

Continuous-Space Simulation

Replace the grid representation with continuous-space movement to more closely approximate actual nest geometry.

Observed Path Comparison

Compare simulated wasp trajectories with observed worker paths to determine which computational strategy most closely resembles real behavior.

Reusable Simulation Library

Refactor the current analysis pipeline into a reusable Python package with separate modules for data loading, scenario generation, agents, strategies, routing, metrics, and benchmarking.

Conclusion

This project demonstrates how real biological observations can be transformed into a reproducible computational routing benchmark.

Using observed nest geometry and feeding-bout data, the project constructs 36 synthetic nest-bout scenarios and evaluates 144 strategy simulations across four movement rules.

The results show a clear advantage for structured route planning.

The TSP strategy achieved:

- The fastest median completion time
- The lowest median movement distance
- The lowest median final larval hunger
- The highest median movement efficiency
- Completion in all 36 benchmark scenarios

The greedy strategy provided the strongest local alternative, while biased and random movement required substantially more travel.

The broader engineering insight is that spatial organization and route planning can have a major effect on agent-based resource-collection efficiency.

The project therefore provides both a biological modeling framework and a general computational benchmark for studying how different movement strategies perform when agents must repeatedly service spatially distributed targets.